MIL-P-46297C (MU)
21 SEPTEMBER 1966
SUPERSEDING
MIL-P-46297B (MU)
30 AUGUST 1963

MILITARY SPECIFICATION

PRIMER, PERCUSSION, M82
METAL PARTS FOR

1. SCOPE

1.1 This specification covers the metal parts for one type of artillery primer designated as Primer, Percussion, M82.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-C-490 — Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.

MILITARY

MIL-A-2550 — Ammunition and Special Weapons; General Specification for.

MIL-P-10025 — Packaging, Packing and Marking for Inert Ammunition Components, General Specification for.

MIL-I-45607 — Inspection Equipment, Supply and Maintenance of.

STANDARDS

FEDERAL

TEST METHOD STANDARD NO. 151 — Metals; Test Methods.

MILITARY

MIL-STD-105 — Sampling Procedures and Table for Inspection by Attributes (ABC-STD-105).

MIL-STD-109 — Quality Assurance Terms and Definitions.

MIL-STD-1168 — Lot Numbering of Ammunition.

MIL-STD-1235 — Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

FSC 1390

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DRAWINGS

ARMY

8861198 — Body

8861200 — Cup, Ignition

8861204 — Container Assembly

8861206 — Closure

8861211 — Plunger

PUBLICATIONS

ARMY

ET-8867962 — Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

3.2 Parts. The parts shall comply with all requirements specified on Drawings (dwgs.) 8861198, 8861200, 8861204, 8861206 and 8861211, as applicable, and with all requirements specified in applicable specifications.

3.3 Mercurous Nitrate test. The primer bodies shall exhibit no cracks when subjected to the mercurous nitrate test specified in 4.4.1.

3.4 Workmanship. All parts shall be dry and free of chips, cracks, dirt, grease, flux, heavy oxide scale and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions specified within this specification.

(b) Number of units of product inspected.

(c) Results obtained, by defect code, for all inspections performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.

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(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 First article inspection.

4.2.1 *Submission.* The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of twenty-five assemblies and twenty-five each of every component and every subassembly which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings for materials, processes and tests. A sample containing known defects will not be submitted unless specifically authorized, by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 *Inspections to be performed.* Assemblies and components will be subjected by the Government to any or all the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

4.2.3 *Rejection.* If any assembly or component fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component to comply with any of the stated requirements.

4.3 Inspection provisions.

4.3.1 *Lot formation.* The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in

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one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105 and shall be numbered in accordance with MIL-STD-1168. Unless otherwise approved by the contracting officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots for components or subassemblies, other than the items of delivery, shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector.

4.3.2 *Examination.* Inspection for Critical defects shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQLs and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect, except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Tube, container (see dwg. 8861203 covering a detail of dwg. 8861204).

Categories	*Defects*	*Method of inspection*	*Code No. (see 6.2)*
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Pitch diameter of thread minimum (min.)	Gage	01001
102.	Major diameter of thread, min	Gage	01002
103.	Length of effective thread, min	Gage	01003
104.	Crack or deformation	Visual	01004
Minor:	AQL 4.00 percent		
201.	Concentricity of outside diameter with threads	Gage	01005
202.	Outside diameter	Gage	01006
203.	Inside diameter of container cup end	Gage	01007
204.	Inside diameter opposite cup end	Gage	01008
205.	Concentricity of inside diameter with thread	Gage	01009
206.	Diameter of thread pilot, min	Gage	01010
207.	Concentricity of the inside diameter of cup end with thread.	Gage	01011
208.	Total length	Gage	01012
209.	Pitch diameter of threads, maximum (max.)	Gage	01013
210.	Major diameter of threads, max	Gage	01014

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Categories	Defects	Method of inspection	Code No. (see 6.2)
211.	Width of slots, min	Gage	01015
212.	Depth of slots	Gage	01016
213.	Depth to interior shoulder, max	Gage	01017
214.	Finish improper	Visual	01018
215.	Evidence of poor workmanship (see 3.4)	Visual	01019

4.3.2.2 Container assembly (see dwg. 8861204).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Cup ruptured	Visual	02001
Minor:	AQL 1.50 percent		
201.	Depth to cup, max	Gage	02002
202.	Braze not 360°	Visual	02003
203.	Evidence of poor workmanship (see 3.4)	Visual	02004

4.3.2.3 Closure (see dwg. 8861206).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor:	AQL 1.50 percent		
201.	Disc ruptured	Visual	03001
202.	Disc not completely fastened to washer	Visual	03002
203.	Washer split	Visual	03003
204.	Evidence of poor workmanship (see 3.4)	Visual	03004

4.3.2.4 Body (see dwg. 8861198).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Concentricity of thread with outside surface	Gage	04001
102.	Concentricity of hole through base with outside surface	Gage	04002

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Categories	Defects	Method of inspection	Code No.
103.	Concentricity of interior surface above hole through base with outside surface	Gage	04003
104.	Concentricity of large inside diameter with outside diameter.	Gage	04004
105.	Pitch diameter of threads, max	Gage	04005
106.	Minor diameter of thread, max	Gage	04006
107.	Thickness of flange	Gage	04007
Minor:	AQL 4.00 percent		
201.	Length to datum diameter	Gage	04008
202.	Thickness from base to bottom of igniter seat	Gage	04009
203.	Concentricity of flange with outside surface	Gage	04010
204.	Thickness from base to shoulder below igniter seat	Gage	04011
205.	Diameter of plunger hole through base	Gage	04012
206.	Diameter of cavity above hole through base	Gage	04013
207.	Diameter of flange	Gage	04014
208.	Large inside diameter	Gage	04015
209.	Pitch diameter, min	Gage	04016
210.	Total length	Gage	04017
211.	Length to thread, max	Gage	04018
212.	Length to end of thread, min	Gage	04019
213.	Finish improper	Visual	04020
214.	Evidence of poor workmanship (see 3.4)	Visual	04021

4.3.2.5 Cup, Ignition (see dwg. 8861200).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Depth of primer cavity	Gage	05001
102.	Diameter of primer cavity	Gage	05002
103.	Concentricity of primer cavity and thread	Gage	05003
104.	Concentricity of flash hole and thread	Gage	05004
105.	Total length	Gage	05005
106.	Flash hole obstructed	Visual	05006
Minor:	AQL 4.00 percent		
201.	Pitch diameter of thread	Gage	05007
202.	Major diameter of thread, min	Gage	05008

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Categories	Defects	Method of inspection	Code No.
203.	Diameter of pilot, max	Gage	05009
204.	Length of pilot	Gage	05010
205.	Diameter of flash hole	Gage	05011
206.	Depth of slot	Gage	05012
207.	Width of slot, min	Gage	05013
208.	Chamfer missing from bottom of primer cavity at mouth of flash hole	Visual	05014
209.	Finish improper	Visual	05015
210.	Evidence of poor workmanship (see 3.4)	Visual	05016

4.3.2.6 Plunger (see dwg. 8861211).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Total length	Gage	06001
102.	Length to shoulder	Gage	06002
103.	Diameter and profile of firing point	Gage	06003
104.	Length of firing point	Gage	06004
105.	Concentricity of diameters	Gage	06005
Minor:	AQL 2.50 percent		
201.	Large diameter	Gage	06006
202.	Small diameter	Gage	06007
203.	Radius or chamfer missing	Visual	06008
204.	Finish improper	Visual	06009
205.	Evidence of poor workmanship (see 3.4)	Visual	06010

4.3.3 *Testing.*

4.3.3.1 *Mercurous nitrate test (see 3.3) — Major defect — Code No. 04022.* Twenty-five bodies shall be randomly selected for this test. If any sample fails to comply with the requirement the lot shall be rejected. The test shall be performed as specified in 4.4.1.

4.3.3.2 *Rupture test (see dwg. 8861204) — Major defect — Code No. 02005.* One hundred and ten container assemblies shall be randomly selected for this test, if any sample fails to comply with the requirement the lot shall be rejected. The test shall be performed as specified in 4.4.2.

4.3.3.3 *Air pressure test (see dwg. 8861204) — Major defect — Code No. 02006.* One hundred and fifty container assemblies shall be randomly selected for this test. If two or more samples fail to comply with the requirement the lot shall be rejected. The

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test shall be performed as specified in paragraph 4.4.3.

4.3.3.4 *Freedom from alkalies and strong acids — Major defect — Code No. 02007.* Three sample containers from each four hours production shall be selected for this test. The test shall be performed as specified in paragraph 4.4.4.

4.3.3.5 *Freedom from oil and grease — Major defect — Code No. 02008.* Three sample containers from each four hours production shall be selected for this test. The test shall be performed as specified in 4.4.5.

4.3.3.6 *Rockwell Hardness test (see dwg. 8861198) — Major defect — Code No. 04023.* Ten bodies shall be randomly selected for this test; if any sample fails to comply with the requirement the lot shall be rejected. The test shall be performed as specified in 4.4.6.

4.3.3.7 *Grain size test (see dwg. 8861198) — Major defect — Code No. 04024.* Ten bodies shall be randomly selected for this test; if any sample fails to comply with the requirement the lot shall be rejected. The test shall be performed as specified in 4.4.7. Primer bodies used in this test shall not be returned to the lot.

4.3.4 *Inspection equipment.* Equipment Tabulation Number ET–8867962 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the ET shall be in accordance with Specification MIL–I–45607.

4.3.4.3 *Government use of contractor's inspection and test equipment.* The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures.

4.4.1 *Mercurous nitrate test.* The primer bodies shall be tested in accordance with Federal Test Method Standard No. 151, Method 831. Each body shall be examined for cracks.

4.4.2 *Rupture test.* The container assembly shall be tested using equipment as specified in 4.3.4.

4.4.3 *Air pressure test.* The sample container shall be tested using equipment as specified in 4.3.4.

4.4.4 *Freedom from alkalies and strong acids.* The container assemblies shall be tested in accordance with Specifications TT–C–490. If the limits specified herein have not been complied with, inspection for acceptance shall be stopped until corrective action has been taken. After corrective action, sampling shall be resumed as specified in 4.3.3.4.

4.4.5 *Freedom from oil and grease.* The container assemblies shall be tested and ac-

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tion taken in accordance with Specification TT C-490.

4.4.6 *Rockwell Hardness test.* The primer bodies shall be tested in accordance with Federal Test Method Standard No. 151, Method 213.1.

4.4.7 *Grain size test.* The primer bodies shall be tested in accordance with Federal Test Method Standard No. 151, Method 312.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.

5.1.1 *Level C.* Packing and marking shall be in accordance with Specification MIL-P-10025.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

(a) Title, number and date of this specification.

(b) Data cards — Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

(c) First article provision.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 *Inspection equipment lists.* Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD — Contractor design on controlled contractor format and/or commercial equipment.

MU — Army design, mandatory for use.

6.3.3 *Army designs.* Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: mandatory (designated "MU") and nonmandatory (indicated by drawing or Federal stock number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with nonmandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabri-

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cation or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL–D–1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 *Submission of contractor designs.* Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND1. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

Custodian:

Army—MU

Preparing activity:

Army—MU

Project No. 1390–A581

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEM PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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